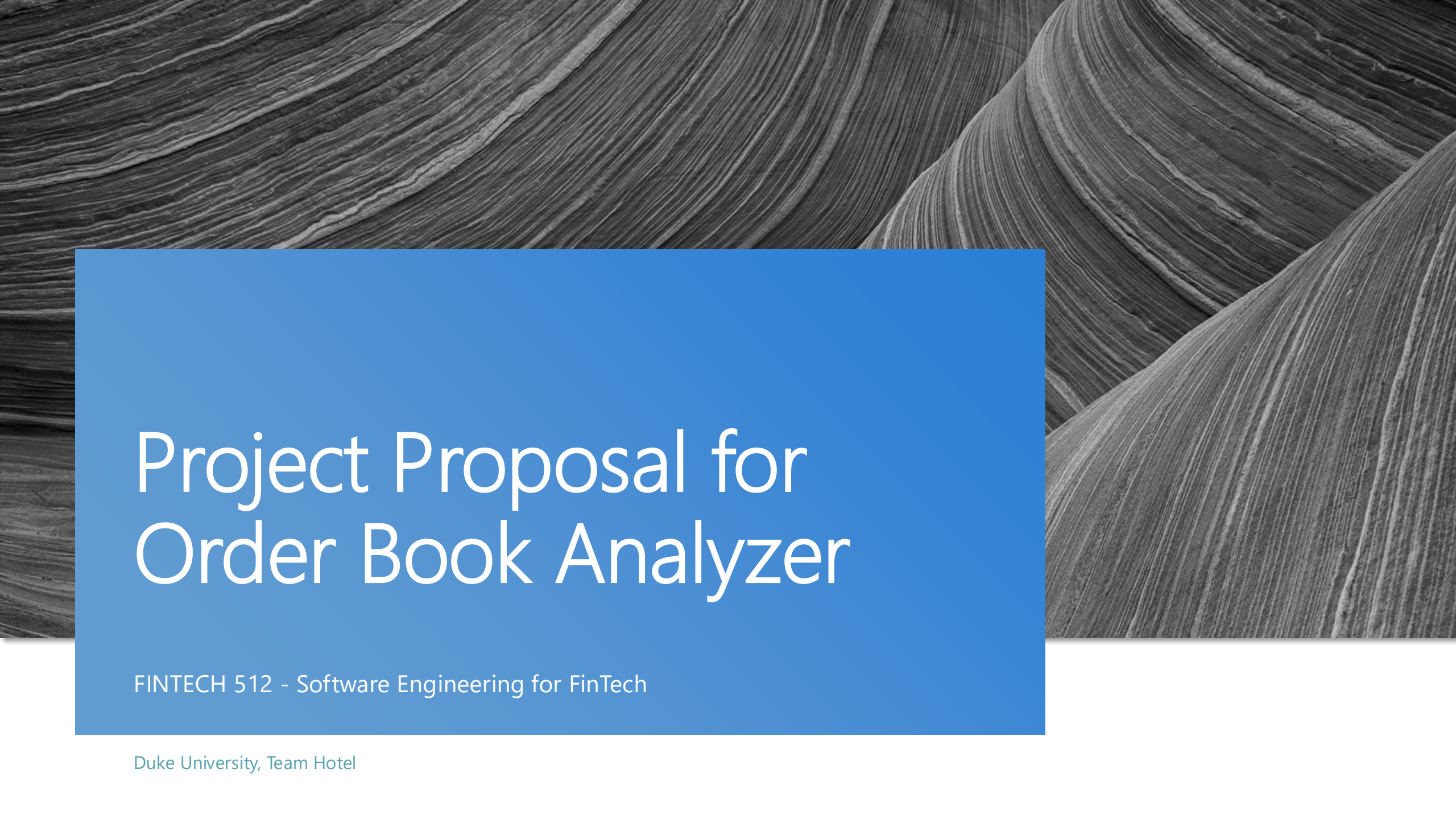




Project Proposal for Order Book Analyzer

FINTECH 512 - Software Engineering for FinTech

Full team of five members for the group project (section 1)



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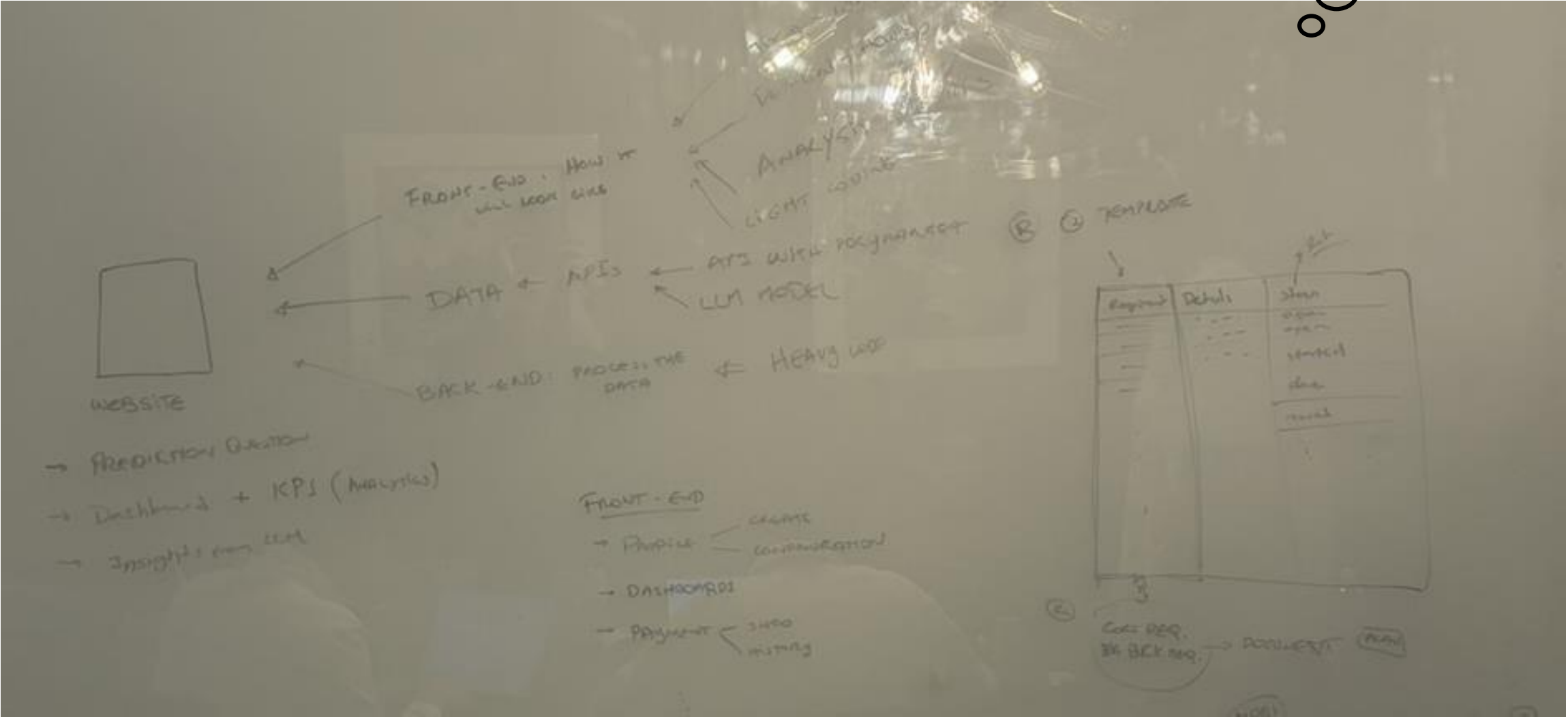


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Agenda

- 01.** Project's 'elevator pitch'
- 02.** Lean Canvas Diagram
- 03.** Paper prototype screens
- 04.** Use case
- 05.** Descriptions of user and administrator roles
- 06.** Data sources
- 07.** Link to your project requirements document

Brainstorming Process



Project's 'elevator pitch'



Bloomberg of Prediction Markets



Prediction markets, where people bet real money on the outcomes of future events (like whether Bitcoin will hit \$110,000 this year), are exploding in popularity. But making sense of all that betting activity can be frustrating. The raw data behind these markets is messy, disorganized, and hard to easily analyze.

Order Book Analyzer cuts through that noise. Think of us as the Bloomberg for prediction markets. We take massive amounts of complicated, second-by-second betting data and translate it into clear, powerful dashboards and insights. Instead of wrestling with messy data, our platform gives you the exact trends and historical information you need to see exactly where the smart money is going.

The Lean Canvas

Designed for:

Order Book Analyzer

Designed by:

Team Hotel

Date:

02/27/2026

Version:

1.0

Problem  List your customer's top 3 problems 1. Access & Opacity: High-frequency data (L2/L3 order books) is notoriously difficult to access, painful data wrangling and gaps. 2. Capacity to Process: Normalizing raw prediction market streams into usable research formats requires specialized engineering effort. 3. Signal-to-Noise Ratio: Traditional financial data is too noisy for pure behavioral signal extraction.	Solution  Outline a possible solution for each problem A website provides Insights & Dashboards: Insights, dashboard, and KPI for one or more popular prediction markets trades	Unique Value Prop.  Single, clear, compelling message that turns an unaware visitor into an interested prospect High quality, tick-level historical data and analytics for high-frequency prediction markets	Unfair Advantage  Can't be easily copied or bought 1. Curated Historical Data Archive: Because live data is fleeting and heavy to store, reconstructing this past data is nearly impossible for new entrants. 2. Domain & Technical Expertise	Customer Segments  List your target customers and users 1. Mid and small size Hedge Funds 2. Sophisticated algo traders who wants to trade in prediction market
Existing Alternatives  List how these problems are solved today. 1. Eagle Alpha 2. Dune Analytics 3. DIY engineering 4. Basic Polymarket Dashboards	Key Metrics  List the key numbers that tell you how your business is doing Growth: 1. # of users 2. Retention 3. API call volumes 4. Daily Active User System Health: 1. Data Quality Score	High-Level Concept  List your X for Y analogy (e.g. YouTube = Flickr for videos) 1. Yahoo Finance for Prediction Market 2. Bloomberg for Prediction Market	Channels  List your path to customers 1. Social Media (LinkedIn, X, Reddit, Red Note) 2. Organic Growth 3. Search Engine 4. Ads	Early Adopters  List the characteristics of your ideal customers. Quants who already engaged with Polymarket who are hitting the limitations of existing visual dashboards and need raw, gapless datasets for rigorous backtesting.
Cost Structure  List your fixed and variable costs. Data Infrastructure: Heavy cloud compute and storage costs (AWS EC2/S3, hosting specialized streaming and database clusters) to maintain gapless WebSocket ingestion. Variable costs for data query and usage. Engineering & People: Time and resources dedicated to pipeline maintenance, API integration, and handling schema changes from the source markets. Customer Acquisition & Marketing: Sales outreach, conference attendance, and content creation. Compliance: Legal frameworks and terms of service management.			Revenue Streams  List your sources of revenue. Revenue Model: Freemium Freemium Model: Basic access to delayed visualizations or aggregated dashboards to drive top-of-funnel user growth. Key Account (KA) Annual Licenses: High-ticket, recurring institutional subscriptions with base data transfer quotas for academic labs and funds. Value-Added Services (Overage): Usage-based billing for massive historical downloads that exceed base quotas, or premium access to specialized datasets.	

Paper prototype screens

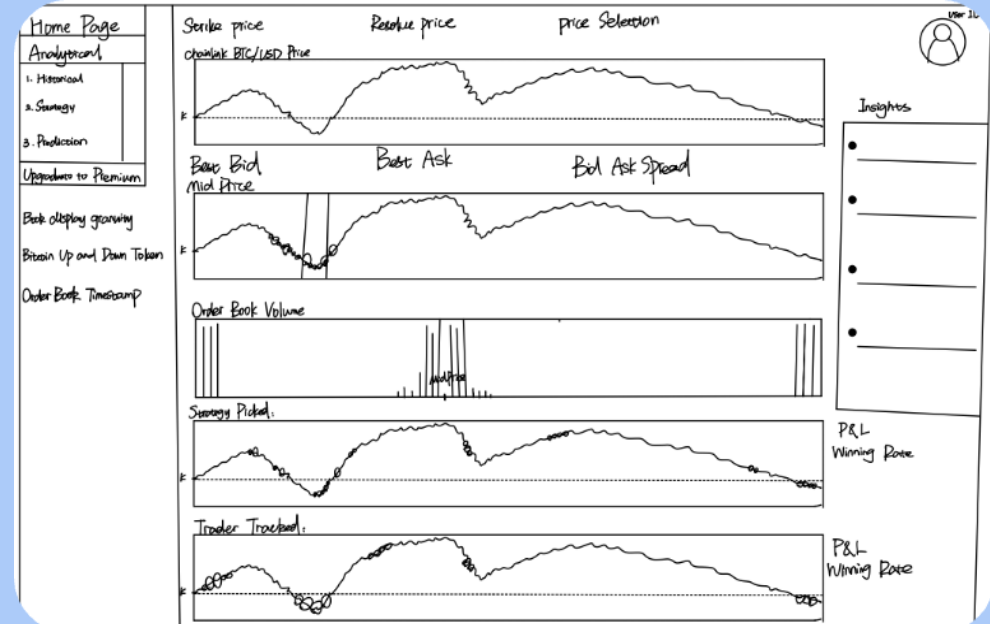
Main Page

Intro page for users to explore



Home Page

Informational dashboard, analytics and KPIs



Use Case: Analyze Bitcoin Prediction Market

Primary Actor:

Retail Bitcoin Investor

Goal: To analyze historical Bitcoin Polymarket data, evaluate back tested strategies, or track top-performing Bitcoin traders

Preconditions:

- The platform's database is populated with current and historical Bitcoin Polymarket data
- The user is authenticated and logged into the system

Main Success Scenario:

User navigates to the main Bitcoin Analysis Dashboard.

System verifies the user's subscription tier (Free Trial vs. Premium).

System automatically renders the dashboard interface, displaying baseline visualization tools for Free users and unlocking advanced analytical tools for Premium users.

User selects one of three primary analysis modules from the interface:

- **Path A (Historical Analytics):** User selects a specific timeframe to view order book snapshots, transaction directions, and price trends.
- **Path B (Strategy Advisory):** User selects a backtested Bitcoin trading strategy to view its performance metrics.
- **Path C (Top-Trader Tracking):** User searches for or selects a specific high-performing trader to view their transaction history and current positions.

System retrieves the requested Bitcoin data for the chosen module and updates the visualizations accordingly.

Alternate Flows (Extensions):

Free Trial Limit Exceeded: If a Free-tier user has exceeded their allotted time or query quota upon loading the dashboard, the system displays a "Usage Limit Reached" overlay, disabling the data modules and prompting an upgrade.

Premium Feature Interaction by Free User: If a Free-tier user attempts to interact with a visually locked Premium feature (e.g., clicking on an advanced predictive model within the Historical Analytics module), the system intercepts the click and displays an upgrade modal.

Descriptions of user and administrator roles

REGULAR USER

The regular users using the analytical data, historical data trends to help with the investing decisions/activities

1. Multi-dimensional data browsing/visualizing: Primarily, users could check the historical price curve, candlestick (k-line), in "5-mins/15-mins" timestamp
2. Percentage progress bar: An intuitive percentage progress bar displays the proportion of funds allocated to "Buy Yes" and "Buy No" orders in the new orders placed within last 5-mins/15mins
3. Check the analytical reports and insights: Display "Best Buy" and "Best Sell" at the certain time periods on dashboard, and some possible suggestions
4. Backtrack historical accuracy: Users could view the final settlement results of the market under similar past trends

ADMINISTRATOR ROLE

The Administrator focus on maintain high frequency data flow, data validation and security

1. Data ingestion and integrity: Ensure uninterrupted data flow in 5-mins/15-mins retrieval from the Polymarket API, clean historical data regularly, and remove abnormal noise caused by API latency to maintain the accuracy and applicability of the real-time data flow
2. System audit: monitor API call frequency to prevent exceeding limits, manage user feedback, and ensure the stable operation of analytics tools
3. Dashboard configuration: Defining the data visualization logics, ensuring the 5-mins/15-mins charts render correctly without lagging

Polymarket Data Service Integration

Sources

Our data service pulls information directly from Polymarket's APIs Official Documentation: [Polymarket API Reference](#)

Following are the target API Endpoints we will leverage:

- **Get Bitcoin Markets:**
`https://gamma-api.polymarket.com/events`
- **Subscribe to Live Order Book Updates (per market):**
`wss://ws-subscriptions-clob.polymarket.com/ws/market`
- **Subscribe to Live Crypto Prices (Chainlink/Binance via Polymarket):**
`wss://ws-live-data.polymarket.com`

Link to your project requirements document

<https://duke.box.com/s/b9j6rdc5zg165nhuga59pv1w49vvsqz5>

Major Requirements:

1. Main Functionality Requirements:

- Enable customers to interact with the website. Specifically, allow customers to create accounts, view data and analytics, and subscribe to the website.
- Maintain running instance of the website, running on a virtual machine to continuously pull data from polymarket and display on the website.
- Enable customers to check their account history and activities.
- Enable users to create new account for themselves. Tests available for developers of the team and TAs

2. Users Requirements:

- Add use cases for new user to access the website, subscribe to the website, view the data, charts, analytics, and check their account.
- Write test cases for checking 2 different types of historical data. e.g.
 - Check bitcoin up market data at 15 minutes duration, by tick, timestamp, December 9, 10:00AM
 - Check bitcoin down market data at 15 minutes, by 1 second buckets, timestamp, December 12, 10:00AM
- Parameters:
 - Market ID: Series number for different types of market
 - Book playback granularity: Duration of the buckets of displayed trades
 - Timestamp: the period of the data which the user wants to investigate.

3. Data Integration Requirements:

- The system shall integrate external data from at least one third-party REST/JSON API (e.g., Polymarket).
- Retrieve data programmatically via HTTP requests.
- Parse and validate JSON responses.
- Persist relevant data in the system database.
- Handle API errors, rate limits, and invalid responses gracefully.

4. Charting Requirements:

- Include the strike price, resolve price, and price at selection
- Include chain-link BTC/USD over time price chart
- Include Best Bid, Best Ask, Spread data
- Include a Mid Price (Snapshot view) over time chart
- Include a size over price chart
- Include cumulative depth over price chart
- Include the trader summary history
- Include a section of loaded data

5. UI/UX Design Focus - Prototyping I: Paper Prototype (Home page/Main page)

- Develop a Home page prototype that tells user what this website is about and display our prop.
- Develop a general data page prototype which will be where the main data charts reside. This will be the core function of this website. It will include all the charts mentioned above, allowing users to pick parameters of their desire and print the corresponding data and chart on the screen with already-calculated analytical indicators.

6. UI/UX Design Focus - Prototyping II: Raw webpages

- Fill in the design of other pages based on the style and core implementation of the main and home pages
- Come up with the overall design scheme, such as background color, font, icon, name, and viewports.

7. UI/UX Design Focus - Prototyping III: Final Polish

- Gather feedback from other peers and see what to improve.
- Make sure the overall style is coherent and consistent
- Make sure it is easy to use, and nothing is hard to see or find.

Project Specific Requirements:

1. System Architecture

- Data Ingestion (Hot Path):
 - Python WebSocket client to ingest market data (PolyMarket).
 - Redis cache for real-time data access.
 - REST/WS strategy execution service for trade decisions.
- Transform Layer
 - Kafka message broker with multiple topics.
 - Debezium for CDC from OLTP database.
 - Python Pub/Sub services for processing and routing events.
- Analytics & Deploy
 - Strategy and backtesting services (Jupyter, Streamlit).
 - Object-oriented strategy design.
 - Containerized deployment (Docker, Docker Compose).
 - CI/CD integration with cloud container registry.
- Persistence (Off Path)
 - OLTP database: PostgreSQL.
 - OLAP database: ClickHouse.

2. Implementation of CI/CD pipeline

- Builds the application on every commit.
- Runs the full automated test suite.
- Executes GitLab SAST scanning.

- Deploys the application to the target VM/container environment upon successful build.

3. Security Requirements

- Threat Modeling:
 - Create a Data Flow Diagram (DFD).
 - Identify assets, actors, and trust boundaries.
 - Perform STRIDE analysis on the system
- Static Analysis
 - Run semgrep on the codebase.
 - Address or document identified vulnerabilities.

4. Architecture Documentation (C4 Model)

- System Context Diagram
- Container Diagram
- Component Diagram
- Code-Level Diagram

5. Logging & Monitoring

- Follow OWASP Logging Cheat Sheet guidelines.
- Define:
 - Events to log
 - Log storage location
 - Log format (structured logging required)
 - Sensitive data handling
- Document in log_design.md
- Log security-relevant events (auth, failures, data access, trade execution).
- Ensure logs support incident investigation.
- Prevent log tampering and unauthorized access.

6. OAuth2 Integration Requirements

- Integrate a third-party OAuth2 provider (e.g., Google).
- Allow users to authenticate via OAuth2 login.
- Securely store and manage access tokens.
- Protect authenticated routes and restrict access to authorized users only.
- Ensure no user passwords are stored locally.

7. Containerization Requirements

- Provide a Dockerfile that builds a production-ready image of the BigBucks application.
- Provide a docker-compose.yml file that:
 - Deploys the application and required services (e.g., database).
 - Creates persistent volumes for database storage.
- Use Docker Compose to deploy and run the application on the target virtual machine.

Thank you!